

25PC2RA202: Sensors and Actuators Laboratory (S&A Lab)

Experiment 9: DC Motor Characterization

Prof. A. Adithya Kashyap, Prof. Harshvardhan Uppaluru

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1. Introduction

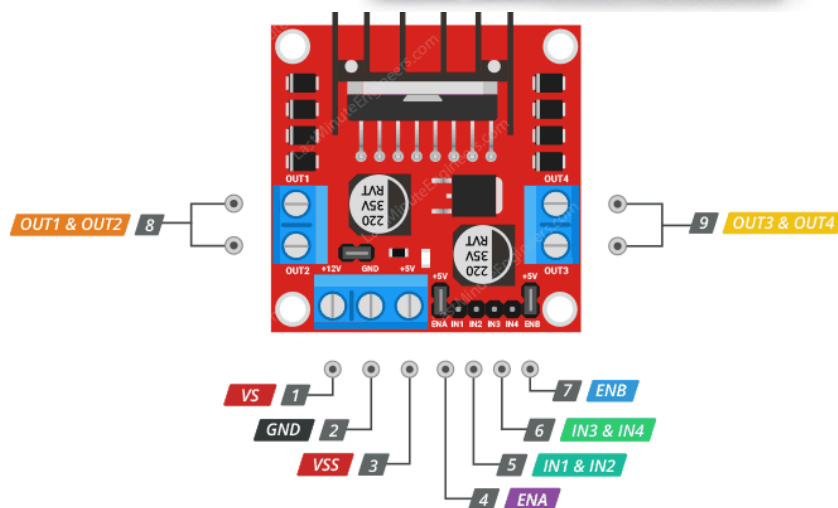
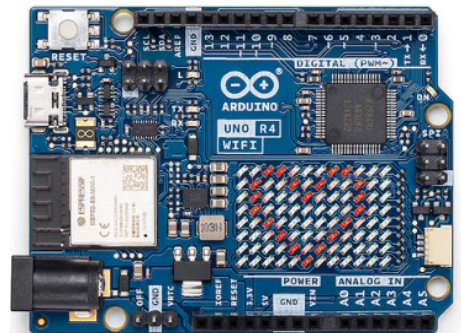
DC motors are the most common type of motors that are used in two-wheel drive robots. These motors are simple, reliable and perfect for beginner robotics projects. They don't need any fancy signals - just apply voltage and they spin.

Why don't we just connect the motors directly to Arduino? DC motors need a lot more current than an Arduino can safely provide. If you try to drive a motor directly from an Arduino pin, you'll risk overloading it and possibly damaging your Arduino.

The L298N motor driver module acts as a bridge between low-power Arduino and high-power motors. Small control signals are sent from the Arduino and L298N safely handles higher current required by the motor. In this experiment, you will learn how to hook up an L298N motor driver to your Arduino, control both the speed and direction of a DC motor.

2. Equipment

- Arduino UNO R4
- L298N motor driver
- DC Motor
- External Power Supply



L298N Module Pinout

Last Minute ENGINEERS.com

3. Arduino Code

```
1 // Motor A connections
2 int enA = 9;
3 int in1 = 8;
4 int in2 = 7;
5
6 void setup() {
7   // Set all the motor control pins to outputs
8   pinMode(enA, OUTPUT);
9   pinMode(in1, OUTPUT);
10  pinMode(in2, OUTPUT);
11
12   // Turn off motor A- Initial state
13   digitalWrite(in1, LOW);
14   digitalWrite(in2, LOW);
15 }
16
17 void loop() {
18   directionControl();
19   delay(1000);
20   speedControl();
21   delay(1000);
22 }
23
24 // This function lets you control spinning direction of motors
25 void directionControl() {
26   // Set motor A to maximum speed
27   digitalWrite(enA, HIGH);
28
29   // Turn on motor A
30   digitalWrite(in1, HIGH);
31   digitalWrite(in2, LOW);
32   delay(2000);
33
34   // Now change motor A direction
35   digitalWrite(in1, LOW);
36   digitalWrite(in2, HIGH);
37   delay(2000);
38
39   // Turn off motor A
40   digitalWrite(in1, LOW);
41   digitalWrite(in2, LOW);
42 }
43
44 // This function lets you control speed of the motors
45 void speedControl() {
46   // Turn on motors
47   digitalWrite(in1, LOW);
48   digitalWrite(in2, HIGH);
49
50   // Accelerate from zero to maximum speed
51   for (int i = 0; i < 256; i++) {
52     analogWrite(enA, i);
53     delay(20);
54   }
55
56   // Decelerate from maximum speed to zero
57   for (int i = 255; i >= 0; --i) {
58     analogWrite(enA, i);
59     delay(20);
60   }
61
62   // Now turn off motor A
63   digitalWrite(in1, LOW);
64   digitalWrite(in2, LOW);
65 }
```

4. Key Concepts - [Link](#)

Using the provided link, study the key concepts to understand motor control:

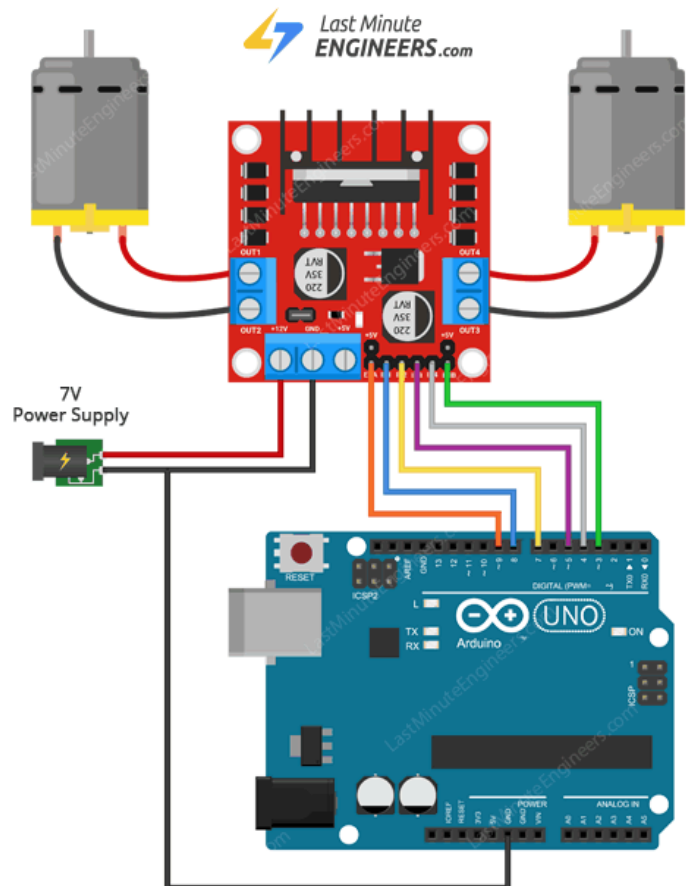
- H-Bridge circuit, and
- Pulse Width Modulation (PWM)

Additionally, study the L298N Motor Driver Chip, L298N Motor Driver Module Pinout, Voltage Drop of L298N and On-board 5V Regulator and Jumper. DO NOT ignore the importance of the concepts defined in the link.

5. Wiring

L298N Pin	Signal	Arduino R4 Pin
VS	Supply Voltage	-
GND	Ground	GND
ENA	PWM	9 (PWM-enabled)
IN1	Input pins	8
IN2	Input pins	7

- In this experiment, you will be using only one DC motor. So you can ignore the connections for the second motor in the wiring diagram on the right side.
- Instead of 7V Power Supply as shown in the wiring diagram, you will be providing 12V Power Supply.



6. Observations

Write the answers to the following questions in your record:

- With ENA HIGH, what did the motor do when IN1 was HIGH and IN2 was LOW?
- What happened when IN1 and IN2 were interchanged (IN1 LOW, IN2 HIGH)? How does the H-bridge reverse the motor without swapping its wires?
- What happened when both IN1 and IN2 were LOW? Why does the motor stop in this condition?
- What did you notice at the moment the motor changed direction? Why can't it reverse instantly?
- If the two motor wires at OUT1 and OUT2 are interchanged, what happens to the direction of rotation for the same input signals? Why?
- What did you observe at low PWM values? Why does the motor hum without rotating?
- In which direction did the motor turn during the speed ramp? Which pins decide the direction, and which pin decides the speed?
- The motor is rated for 12 V and the L298N is supplied with 12 V. Does the motor run at full speed?

7. Evaluation Rubric

Assessment Component	Marks	Awarded Marks
Wiring	3	
Programming	2	
Observations	4	
Attendance	1	
Total	10	

Signatures

Prof. A. Adithya Kashyap

Date:

Prof. Harshvardhan Uppaluru

Date: